Shreya Chivilkar

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Education

Georgia Institute of Technology, Atlanta, GA **August 2024 – May 2026**
Master of Science in Computer Science *CGPA : 4.0/4.0*
Coursework: Advanced Programming Techniques, Algorithms, GPU Programming, Systems for Machine Learning, Data & Visual Analytics, Enterprise Cybersecurity Management, Machine Learning, Computer Vision, AI for Social Impact

Veermata Jijabai Technological Institute, Mumbai, India **August 2017 – May 2021**
Bachelor of Technology in Electronics and Telecommunication Engineering *CGPA : 9.45/10.0*
Coursework: Computer Programming (C++, Java), Operating Systems, Database Management Systems, Computer Architecture

Technical Skills

Programming Languages: C++, Python, Java, SQL
Framework & Tools: CMake, Spring Boot, FastAPI, Flask, Node.js, React.js, Kafka, Solace, PlantUML, Streamlit, Tableau, JIRA
Machine Learning: PyTorch, TensorFlow, Scikit-Learn, BERT, Federated Learning, Causal ML, CUDA
Cloud & DevOps: AWS (EC2, RDS, S3), Docker, Kubernetes, Openshift, Jenkins, Git, GitLab, JIRA, Bitbucket, Perforce
Databases: Oracle DB, MongoDB, Redis, Elasticsearch, Kibana

Work Experience

MathWorks, Software Development Intern **May 2025 – August 2025**
Backend Systems | ML-Driven Code Analysis *Massachusetts, United States*

- Built an interactive analysis tool in **Python** for shared object (.so) files, mapping **700+ function symbols** across **60+ DLLs** and applying **TF-IDF** and **Agglomerative clustering** to identify modular refactoring opportunities.
- Engineered a graph partitioning framework using **Louvain** and **Kernighan–Lin algorithms**, splitting large components into cohesive modules and reducing cross-component coupling by **30%**.
- Extended an internal Node.js/React "Callsite Risk Analysis" dashboard for tracking dynamic name-resolution conflicts across large codebases; built risk-aware search and filtering APIs and automated JIRA-based ticket routing to reduce manual triage effort.

Citibank, Software Development Engineer II **July 2023 – July 2024**
Microservices | Real-time Systems | Agile Methodologies

- Built a real-time trader dashboard using **Java, Spring Boot** with **35+ APIs**, integrating WebSockets & Kafka to stream intraday exposure and P&L. Reduced latency from **22s to 400ms** by migrating high-volume SQL queries to Elasticsearch.
- Reduced fat-finger and mis-order incidents by **37%** across **7 asset classes** by leading end-to-end development of a **What-If trade simulation system**, delivering risk-mitigation features across the **SDLC** to enable traders to preview outcomes before execution.
- Developed **AskAL**, an automated request-routing system in Spring Boot using **NLP techniques** to process **1K+ client emails** and trigger backend API workflows, reducing manual query handling by **67%**; deployed via **CI/CD pipelines** (Jenkins, Docker).
- Mentored 10+ interns on coding best practices, system design reviews and debugging production issues, accelerating onboarding and engineering quality; also contributed to 20+ technical interviews for backend roles

Citibank, Software Development Engineer I **July 2021 – July 2023**
Distributed Systems | Event-Driven Backends | System Reliability

- Designed and built a **low-latency, distributed** reconciliation and replay service using Java, Ambrosia, and event streaming to detect data inconsistencies and enable deterministic recovery, improving reliability of high-volume financial data pipelines.
- This service delivered robust trade recovery and was adopted across Citi's Global Electronic Brokerage teams as a **plug-and-play solution**, reducing annual costs by **35%**.
- Developed a **scalable, event-driven** subscription platform using **Spring Boot, Kafka**, and the Observer Pattern to automate personalized report delivery for **5K+ clients**, reducing manual dashboard interactions by **55%** and streamlining trader workflows.

Projects

Async Federated Fine-Tuning (FwdLLM) | ***Systems for AI Lab*** **Aug 2025 – Present**

- Improved federated fine-tuning throughput by **25%** by mitigating straggler effects via an asynchronous training design, reducing idle GPU time by **30%** in a distributed training setup.
- Designed a clean, modular training framework enabling controlled comparisons of synchronous vs. asynchronous execution, incorporating staleness-weighted updates and Oort-style client selection to preserve model quality at scale.

JobBuddy, Morgan Stanley Code to Give Hackathon | **2nd Prize** ***GitHub*** **Sep 2024**

- Built **JobBuddy**, an AI resume–JD matching platform with a **Python (Flask) REST backend**, using DistilBERT embeddings and cosine similarity to match candidates to roles; achieved **87% match accuracy** with **<200ms API latency**.
- Implemented resume auto-parsing and automated email notification workflows, using **Firebase** for persistent storage and backend triggers to support a smooth end-to-end candidate–recruiter experience.

Publications and Achievements

- **Publications:** J. Ben Tamo, N. S. Chouhan, M. C. Nnamdi, Y. Yuan, **S. S. Chivilkar**, W. Shi, S. W. Hwang, B. R. Brenn, M. D. Wang (2025), **"Causal Machine Learning for Surgical Interventions"**, *Big Data Mining and Analytics*, *link* | H. Rajasekharan, S. Chivilkar, N. Bramhankar (2021), **"EEG-based Mental Workload Assessment using a Graph Attention Network"** 2021 IEEE 20th International Conference on Cognitive Informatics & Cognitive Computing *link*
- **Hackathons:** Secured 2nd Place at the Morgan Stanley Code to Give Hackathon Alpharetta (2024) | People's Choice Award at the Markets Tech Hackathon, Citi | 1st Place at the Geeks+ Hackathon, Citi | Qualified for Smart India Hackathon - Stage 2